# Formal Review Template

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Reviewer Name: Justin Pedrozo
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Review Type (e.g., code, usability, design, other): Code

## Scope:

Artifact Reviewed:
questionnaire_controller.rb

p_questionnaire.yml

f_questionnaire.yml

s_questionnaire.yml

careers.yml

Review Objectives:
Ensure that the code works as intended.

## Methodology:

Approach Taken (e.g., execute, inspect, observe): Inspect

Evaluation Criteria (e.g., expected behavior, best practices, heuristic evaluation): Expected behavior and best practices.

Coverage / Limitations (e.g., test login only, ignore database integration): Controller file and Yml files only

## Findings

**Strengths:**

- There is a clear separation of the configuration from logic
- Overall Design is solid: configuration-driven, phased, normalized, and persisted thoughtfully.

**Issues / Defects:**

| Description of | Severity (Low/Med/High) | Evidence / | Suggested |
|---|---|---|---|

| Issue | | Example | Fix |
|---|---|---|---|
| Operator precedence error in `d_index` calculation causes index to land out of bounds, skipping the final phase | High | `d_index = index - @p_questions.length + filtered_s_questions.length` evaluates as `(index - p) + s` instead of `index - (p + s)` | Add parentheses: `d_index = index - (@p_questions.length + filtered_s_questions.length)` |
| `FQuestion` struct arity mismatch because it is defined with 3 fields but constructed with 5 | High | `FQuestion = Struct.new(:id, :prompt, :career)` vs. `FQuestion.new(i, f["prompt"], f["career"], f["pointer"], f["previous"])` | Add the missing fields to struct |
| | | | |